Multivariate Regression and Classification

M514: Econometrics II

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Multivariate Regression

Recall: Univariate regression model:

$$Y = X'\beta + e$$

where $Y \in \mathbb{R}, X \in \mathbb{R}^k, e \in \mathbb{R}$. Multivariate regression is a system of m linear regressions written as

$$Y_j = X'\beta_j + e_j \tag{1}$$

where $Y_j \in \mathbb{R}, X \in \mathbb{R}^k, e_j \in \mathbb{R}$ for $j = 1, \ldots, m$. Here, the subscript j denotes the j-th dependent variable, not the i-th individual. We model m dependent variables $(Y_j)_{j=1}^m$ using a common vector $X \in \mathbb{R}^k$.

Multivariate Regression: Vector form

The most common way to write the multivariate regression in a vector form is given as

$$\underset{1\times m}{Y'} = \underset{1\times k}{X'}\underset{k\times m}{B} + \underset{1\times m}{e'} \tag{2}$$

where Y, e, B are now given as

$$Y = \begin{pmatrix} Y_1 \\ \vdots \\ Y_m \end{pmatrix}, e = \begin{pmatrix} e_1 \\ \vdots \\ e_m \end{pmatrix}, B = \begin{pmatrix} \beta_1 & \dots & \beta_m \end{pmatrix}$$

Here, the j-th element of Y' in (2) denotes (1).

Multivariate Regression: Matrix form

Let $\mathbf{Y} \in \mathbb{R}^{n\times m}, \mathbf{X} \in \mathbb{R}^{n\times k}$ be matrices collecting each samples in the rows, $\mathbf{e} \in \mathbb{R}^{n\times m}$ be the matrix collecting the error terms.

$$\mathbf{Y} = \begin{pmatrix} Y_1' \\ \vdots \\ Y_n' \end{pmatrix}, \mathbf{Y} = \begin{pmatrix} X_1' \\ \vdots \\ X_n' \end{pmatrix}, \mathbf{e} = \begin{pmatrix} e_1' \\ \vdots \\ e_n' \end{pmatrix}$$

Now, Y_i' denotes the i-th sample of Y', not the j-th covariate of Y'. The multivariate regression in matrix form can be written as

$$\mathbf{Y} = \mathbf{X}B + \mathbf{e}$$

Multivariate Regression: Estimation

As in univariate case, we use the least squares estimator. In other words, we find B that minimizes the size of the squared error.

$$\mathbf{Y} - \mathbf{X}B$$

In the univariate case, the error vector was n dimensional vector and we minimized the Euclidean norm. Now, the error matrix is given by $n \times m$ dimensional matrix and we minimize the Frobenius norm given by

$$\sqrt{\mathsf{tr}\left((\mathbf{Y} - \mathbf{X}B)'(\mathbf{Y} - \mathbf{X}B)\right)}$$

Multivariate Regression: Estimation

Since minimizer does not change after taking square, the estimate B is given as

$$\hat{B} = \text{arg}\min_{B} \text{tr}\left((\mathbf{Y} - \mathbf{X}B)'(\mathbf{Y} - \mathbf{X}B)\right)$$

Using the definition of the trace, we see that

$$\begin{aligned}
\text{tr}\left((\mathbf{Y} - \mathbf{X}B)'(\mathbf{Y} - \mathbf{X}B)\right) &= \sum_{j=1}^{m} \left[(\mathbf{Y} - \mathbf{X}B)'(\mathbf{Y} - \mathbf{X}B)\right]_{jj} \\
&= \sum_{j=1}^{m} (\mathbf{Y}_j - \mathbf{X}\beta_j)'(\mathbf{Y}_j - \mathbf{X}\beta_j)
\end{aligned}$$

where $\mathbf{Y}_j$ is j-th column of $\mathbf{Y}$ and β_j is the j-th column of B.

Multivariate Regression: Estimation

Thus, the joint minimization problem with respect to B splits into the minimization of $(\mathbf{Y}_j - \mathbf{X}\beta_j)'(\mathbf{Y}_j - \mathbf{X}\beta_j)$ with respect to each of β_j.

$$\begin{aligned}
\min_B \operatorname{tr}\big((\mathbf{Y} - \mathbf{X}B)'(\mathbf{Y} - \mathbf{X}B)\big) &= \min_B \sum_{j=1}^{m} (\mathbf{Y}_j - \mathbf{X}\beta_j)'(\mathbf{Y}_j - \mathbf{X}\beta_j) \\
&= \sum_{j=1}^{m} \min_{\beta_j} (\mathbf{Y}_j - \mathbf{X}\beta_j)'(\mathbf{Y}_j - \mathbf{X}\beta_j)
\end{aligned}$$

This means that the multivariate regression just performs equation-by-equation univariate least squares.

Multinomial Logit (Multiple Choice) Model

Let Y be a multinomial random variable that takes values in a finite set $\{1, 2, \ldots, J\}$. We are interested in estimating the conditional probability

$$P_j(x) = \mathbb{P}\{Y = j | X = x\}$$

where $X \in \mathbb{R}^k$ are the predictors thought to explain randomness in Y.

Multinomial Logit (Multiple Choice) Model

Suppose the label $\{1, \ldots, J\}$ does not have a natural ordering. What would be a natural way of modeling $Y \in \{1, \ldots, J\}$?
Let y_j be an indicator function that takes value one if $Y = j$, i.e.,

$$y_j = \begin{cases} 1 & \text{if } Y = j \\ 0 & \text{if } Y \neq j \end{cases}$$

Then, $Y \in \{1, \ldots, J\}$ may be expressed as a J dimensional vector

$$y = \begin{pmatrix} y_1 \\ \vdots \\ y_J \end{pmatrix}$$

Note that each element in the vector y takes values in $\{0, 1\}$ and sum up to one.

Multinomial Logit (Multiple Choice) Model

For each $j = 1, \dots, J$, suppose that we model the conditional probability using logit function

$$\mathbb{P}\{Y = j | X = x\} = \mathbb{P}\{y_j = 1 | X = x\} = \frac{1}{1 + e^{-x'\beta_j}} = \frac{e^{x'\beta_j}}{1 + e^{x'\beta_j}} \tag{3}$$

How do we construct the likelihood of this model?

Multinomial Logit (Multiple Choice) Model

The joint probability mass function is

$$p(Y|X,\beta) = \prod_{j=1}^{J} \mathbb{P}\{Y = j|X,\beta_j\}^{y_j} = \prod_{j=1}^{J} \left(\frac{e^{X'\beta_j}}{1+e^{X'\beta_j}}\right)^{y_j}$$

and the log likelihood is given by

$$\begin{aligned}\log \prod_{i=1}^{n} p(Y_i|X_i,\beta) &= \sum_{i=1}^{n} \log p(Y_i|X_i,\beta) = \sum_{i=1}^{n} \log \prod_{j=1}^{J} \left(\frac{e^{X_i'\beta_j}}{1+e^{X_i'\beta_j}}\right)^{y_{ji}} \\ &= \sum_{i=1}^{n}\sum_{j=1}^{J} y_{ji} \log\left(\frac{e^{X_i'\beta_j}}{1+e^{X_i'\beta_j}}\right) = \sum_{j=1}^{J}\left(\sum_{i=1}^{n} y_{ji} \log\left(\frac{e^{X_i'\beta_j}}{1+e^{X_i'\beta_j}}\right)\right)\end{aligned}$$

where y_{ji} is the i-th sample of y_j.

Multinomial Logit (Multiple Choice) Model

Notice that

$$\begin{aligned}\max_{\beta} \log \prod_{i=1}^{n} p(Y_i|X_i,\beta) &= \max_{\beta} \sum_{j=1}^{J} \left(\sum_{i=1}^{n} y_{ji} \log \left(\frac{e^{X_i'\beta_j}}{1+e^{X_i'\beta_j}} \right) \right) \\ &= \sum_{j=1}^{J} \max_{\beta_j} \left(\sum_{i=1}^{n} y_{ji} \log \left(\frac{e^{X_i'\beta_j}}{1+e^{X_i'\beta_j}} \right) \right)\end{aligned}$$

Thus, the joint maximization problem splits into individual maximization problems where we maximize over $\beta_j, j = 1, \ldots, J$ separately. This means that we may estimate each of $\beta_j, j = 1, \ldots, J$ separately by maximizing the log-likelihood constructed using each of $(y_j, X), j = 1, \ldots, J$. This is equation-by-equation binary logit model. (Recall how we estimated the multivariate regression model!)

Multinomial Logit (Multiple Choice) Model

One problem with this method is that the probabilities do not sum up to one. We want the property

$$\sum_{j=1}^{J} \mathbb{P}\{Y = j | X = x\} = 1$$

To enforce this constriant, we modify the equation (3) as

$$\begin{aligned} \mathbb{P}\{Y = j | X = x\} &= \frac{e^{x'\beta_j}}{1 + e^{x'\beta_j}} \\ \implies \mathbb{P}\{Y = j | X = x\} &= \frac{e^{x'\beta_j}}{\sum_{k=1}^{J} e^{x'\beta_k}} \end{aligned}$$

Now, this model is not estimable by equation-by-equation binary logit model because the solution β_j^* is coupled together with the solution β_k^* for all $k \neq j$.

Multinomial Logit (Multiple Choice) Model

Notice that the joint probability mass function is

$$
\begin{aligned}
p(Y|X,\beta) &= \prod_{j=1}^{J} \mathbb{P}\{Y=j|X\}^{y_j} \\
&= \prod_{j=1}^{J} \left(\frac{e^{X'\beta_j}}{\sum_{k=1}^{J} e^{X'\beta_k}} \right)^{y_j}
\end{aligned}
$$

and the log likelihood is given by

$$
\begin{aligned}
\log \prod_{i=1}^{n} p(Y_i|X_i,\beta) = \sum_{i=1}^{n} \log p(Y_i|X_i,\beta) &= \sum_{i=1}^{n} \log \prod_{j=1}^{J} \left(\frac{e^{X_i'\beta_j}}{\sum_{k=1}^{J} e^{X_i'\beta_k}} \right)^{y_{ji}} \\
&= \sum_{i=1}^{n} \sum_{j=1}^{J} y_{ji} \log \left(\frac{e^{X_i'\beta_j}}{\sum_{k=1}^{J} e^{X_i'\beta_k}} \right)
\end{aligned}
$$

where y_{ji} is the i-th sample of y_j. We jointly maximize this with respect to all $\beta_j, j = 1, \ldots, J$.